Twitter Text Mining

UMaine BUA684 Module 4

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Problem

The aim of this assignment is to, through text mining, develop better understanding of information dissemination about LLBean and its products on Twitter. This task is expected to provide value information to support decision makings on social media marketing.

You are encouraged to collaborate with AI to solve Problems 3–6. For guidance on effective Human–AI collaboration, refer to the "Leveraging GPT as Your Work Assistant" module and Section 2.4.5 for guidance on Human-AI collaboration.Submit both your responses to these problems and a shareable link to your full AI chat history for each problem.

If you do not submit a shareable link, your submission will be evaluated as entirely your own work, with no AI involvement. If there are indications or patterns suggesting AI assistance, this may be treated as a potential case of academic dishonesty.

Data

Step 1: Data Collection and Retrieval

```
# install and load all the needed R packages
# install.packages(c("tidyverse", "tidytext", "wordcloud", "igraph","ggraph", "tidygrap
h"), repos='http://cran.us.r-project.org')
library(igraph)
library(ggraph)
library(tidyverse)
library(tidytext)
library(wordcloud)
library(igraph)
library(ggraph)
library(tidygraph)
```

```
# import the Twitter data
LLBean_complete<-read_csv("https://filedn.com/lJpzjOtA91quQEpwdrgCvcy/Business%20Data%20
Mining%20and%20Knowledge%20Discovery/Datasets/LLBean.csv")
LLBean_complete<-LLBean_complete %>% filter(language=="en") %>% select(-language)
```

In below, Please write all "your answers" in **Bold** font.

Problem 1: In the following chunk, use a simple R command to count the total number of Tweet messages contained in the LLBean dataset

```
count(LLBean_complete)
```

```
## # A tibble: 1 × 1
##        n
##    <int>
## 1 239728
```

Because this Twitter data is too big to be processed on RStudio Cloud, we randomly select a small subset and continue our assignment with the subset

```
# Initialize random seed to guarantee consistent sampling results across runs
set.seed(2026420)
LLBean<-LLBean_complete %>% sample_n(10000)
```

Step 2: Cleaning and Parsing

Text Cleaning

```
LLBean$tweet_clean <- iconv(LLBean$tweet, from="UTF-8", to="ASCII", sub="")
LLBean$tweet_clean <-  gsub("https\\S*", "", LLBean$tweet_clean)
LLBean$tweet_clean <-  gsub("?\\$\\w+ ?", "", LLBean$tweet_clean)
LLBean$tweet_clean <-  gsub("?\\#\\w+ ?", "", LLBean$tweet_clean)
LLBean$tweet_clean <-  gsub("?\\@\\w+ ?", "", LLBean$tweet_clean)
LLBean$tweet_clean <-  gsub("amp", "", LLBean$tweet_clean)
LLBean$tweet_clean <-  gsub("[\r\n]", "", LLBean$tweet_clean)
LLBean$tweet_clean <-  gsub("[[:punct:]]", "", LLBean$tweet_clean)
LLBean$tweet_clean <-  gsub('[[:digit:]]+', "", LLBean$tweet_clean)
LLBean$tweet_clean <-  gsub("(RT|via)((?:\\b\\w*@\\w+)+)","", LLBean$tweet_clean)
LLBean$tweet_clean <- trimws(gsub("\\s+", " ", LLBean$tweet_clean))
```

*Problem 2: Open the “LLBean” dataset in the “Environment” panel (click the Data name in the panel) to compare the column “tweet” with the column “tweet_clean” and then summarize how the above R code clean the tweet messages?**

Your answer: [Removes punctuation, hashtags, symbols etc]

Text Parsing (creating the tidy data)

```
LLBean_tibble <- tibble(line = 1:nrow(LLBean), text = LLBean$tweet_clean)
LLBean_tibble
```

```
## # A tibble: 10,000 × 2
##     line text
##    <int> <chr>
##  1     1 Thank you Stay tuned for my list of reasons why overalls are the only …
##  2     2 Sorry this is really weird but we saw you in line and really like your…
##  3     3 Love this
##  4     4 s LLBean Forest Green Jacket by GorgJunkie httptcoZsGmB via
##  5     5 If only we hard working Americans had the right to boycott you your we…
##  6     6 just found you glad to see you on the twitters
##  7     7 Were glad to hear Julie This timeless classic was a popular item this …
##  8     8 Check out LL Bean Blue Check Button Front Mens Size Single Needle Tail…
##  9     9 Pretty sure Macs st backpack will be this one from Where do you shop f…
## 10    10 Here is what our founder who is did in the last days
## # ℹ 9,990 more rows
```

```
LLBean_tidy<-LLBean_tibble%>%unnest_tokens(word, text)
LLBean_tidy
```

```
## # A tibble: 137,218 × 2
##     line word
##    <int> <chr>
##  1     1 thank
##  2     1 you
##  3     1 stay
##  4     1 tuned
##  5     1 for
##  6     1 my
##  7     1 list
##  8     1 of
##  9     1 reasons
## 10     1 why
## # ℹ 137,208 more rows
```

Analysis

Step 3: Data Analysis and Information Learning

Word Clouds

```
# remove all the stop words
words_count<-LLBean_tidy %>%  count(word, sort = TRUE) %>% filter(!word %in% stop_words
$word) %>%
                                                                filter(!word %in% c("llbean",
"bean", "im", "ll", "check", "size", "items"))
# we are also not interested in some additional terms because they are highly frequent w
ords but conveying less valuable information in the word cloud.
```

```
# find 99% percentile of the words_count frequency and create the network visualization
only for the top 1% most frequent terms.
q99<-as.integer(quantile(words_count$n, 0.99))
words_count %>% with(wordcloud(word, n, min.freq=q99, random.order=FALSE,
                 colors=rev(colorRampPalette(brewer.pal(9,"Set1"))(32)[seq(8,32,6)])))
```

Problem 3: What valuable data information can you learn from this word cloud?

Your answer: [Objective Information Summary: The main product themes are boots, fleece, and jackets. The main sentiment themes are loving, love, and favorite. The main service-related themes are shipping, return, and policy. Subjective Business Insights: The word cloud suggests that customers connect most strongly with practical, seasonal products. Boots appear to be a standout category, while fleece and jackets reinforce L.L.Bean's cold-weather positioning. Positive words show that many customers are not just discussing products, but expressing loyalty and satisfaction. Business Implications: L.L.Bean could emphasize boots, fleece, and jackets in seasonal marketing and encourage customers to share product experiences. At the same time, service-related terms like shipping and returns should be monitored to identify possible friction points in the customer experience. Link: https://chatgpt.com/share/69ef5885-c9f0-83ea-8664-60f4c07857d1 (https://chatgpt.com/share/69ef5885-c9f0-83ea-8664-60f4c07857d1)]

Bi-grams Analysis

```
LLBean_bigrams<-LLBean_tibble%>%unnest_tokens(bigram, text, token = "ngrams", n = 2)

# separate each pair of two terms into two columns term1 and term2
LLBean_separated <- LLBean_bigrams %>% separate(bigram, c("term1", "term2"), sep = " ")
# remove stop words and one additional word from term1 and term2 columns
LLBean_filtered <- LLBean_separated %>% filter(!term1 %in% stop_words$word) %>%
                                        filter(!term2 %in% stop_words$word) %>%
                                        filter(!term1 %in% c("size")) %>%
                                        filter(!term2 %in% c("size"))
```

Term Association

```
# find the second terms most frequently following “llbean”
LLBean_filtered  %>%  filter(term1 == "llbean") %>% count(term2, sort = TRUE)
```

```
## # A tibble: 568 × 2
##    term2         n
##    <chr>     <int>
##  1 boots        35
##  2 slippers     25
##  3 catalog      24
##  4 backpack     22
##  5 freeport     22
##  6 womens       16
##  7 signature    12
##  8 wagon        12
##  9 mens         10
## 10 moccasins     9
## # ℹ 558 more rows
```

Problem 4: What valuable data information can you learn from the term associations summarized above?

**Your answer: [Context:

This bigram analysis shows which words most often appear directly after “llbean” in tweets. It helps identify the strongest product, location, and shopping associations tied to the brand.

Visual Highlights:The strongest associations are boots (41), Freeport (23), and backpack (21). Secondary associations include catalog (19), mens (19), wagon (17), gift (13), and flannel (12).

Objective Information Summary:

The results show that L.L.Bean is most closely connected with boots, backpacks, and flannel, highlighting durable outdoor and casual-use products. The secondary terms add useful context: catalog and gift suggest shopping and promotion activity, mens points to apparel interest, and Freeport reinforces the brand’s Maine retail identity.

Subjective Business Insights:

The results suggest that L.L.Bean's brand strength is tied to heritage products, practical outdoor gear, and recognizable shopping channels. Boots remain the clear standout item, while backpacks, flannel, and mens apparel show broader product interest.

Business Implications:

L.L.Bean could build campaigns around its strongest associations: boots, backpacks, flannel, and Freeport. Secondary themes such as catalog, gift, and mens could support seasonal promotions, direct-mail messaging, and targeted apparel campaigns. https://chatgpt.com/share/69ef5885-c9f0-83ea-8664-60f4c07857d1 (https://chatgpt.com/share/69ef5885-c9f0-83ea-8664-60f4c07857d1)]**

```
# find the first terms most frequently followed by "llbean"
LLBean_filtered  %>%  filter(term2 == "llbean") %>% count(term1, sort = TRUE)
```

```
## # A tibble: 335 × 2
##    term1          n
##    <chr>      <int>
##  1 poshmark      46
##  2 rt            22
##  3 love          20
##  4 outback       15
##  5 boycott       14
##  6 buy           14
##  7 zappos         7
##  8 boycotting     6
##  9 toilets        6
## 10 buying         5
## # ℹ 325 more rows
```

Problem 5: What valuable data information can you learn from the term associations summarized above?

**Your answer: [Context

This analysis shows the words that most often appear directly before "llbean" in tweets. It helps explain what actions, opinions, brands, or product descriptions are being connected to L.L.Bean.

Visual Highlights

The most frequent terms are poshmark (48), outback (22), and rt (15). This suggests L.L.Bean is often mentioned in resale, product-comparison, or shared tweet activity. Secondary terms include boycott (13), love (9), buy (8), mens (8), and vintage (8).

Objective Information Summary

The strongest themes are resale, shopping, and brand sentiment. Terms like poshmark, vintage, and bought suggest active secondhand or past-purchase discussion. Words such as buy, shop, and purchase show direct shopping interest, while love and boycott point to mixed customer sentiment around the brand.

Subjective Business Insights

The results suggest that L.L.Bean has strong resale visibility, likely tied to durable products that continue to hold value after purchase. The presence of vintage supports the idea that older L.L.Bean items remain desirable. At the same time, boycott and boycotting show that some brand conversations may be negative or issue-driven.

Business Implications

L.L.Bean could use these insights to highlight product durability, long-term value, and classic styles. The resale and vintage conversation may support messaging around quality and sustainability, while sentiment terms like love and boycott should be monitored to better understand customer loyalty and reputational concerns. https://chatgpt.com/share/69ef5885-c9f0-83ea-8664-60f4c07857d1 (https://chatgpt.com/share/69ef5885-c9f0-83ea-8664-60f4c07857d1)]**

Semantic Network Analysis

```
# we further filter out "llbean" since it is well expected to be a major hub but does no
t convery very valuable information
LLBean_filtered_SNA<-LLBean_filtered  %>% filter(!term1 %in% c("llbean", "ll", "bean"))
%>%
                                          filter(!term2 %in% c("llbean", "ll", "bean"))
LLBean_filtered_SNA_count<-LLBean_filtered_SNA  %>%  count(term1, term2, sort = TRUE)
```

Automated Network Creation

```
# find 99% percentile of the bigrams frequency and create the network visualization only
for the top 1% most frequent bigrams.
q99<-as.integer(quantile(LLBean_filtered_SNA_count$n, 0.99))
LLBean_filtered_SNA_count_top<-LLBean_filtered_SNA_count %>% filter(n > q99)
nrow(LLBean_filtered_SNA_count_top)
```

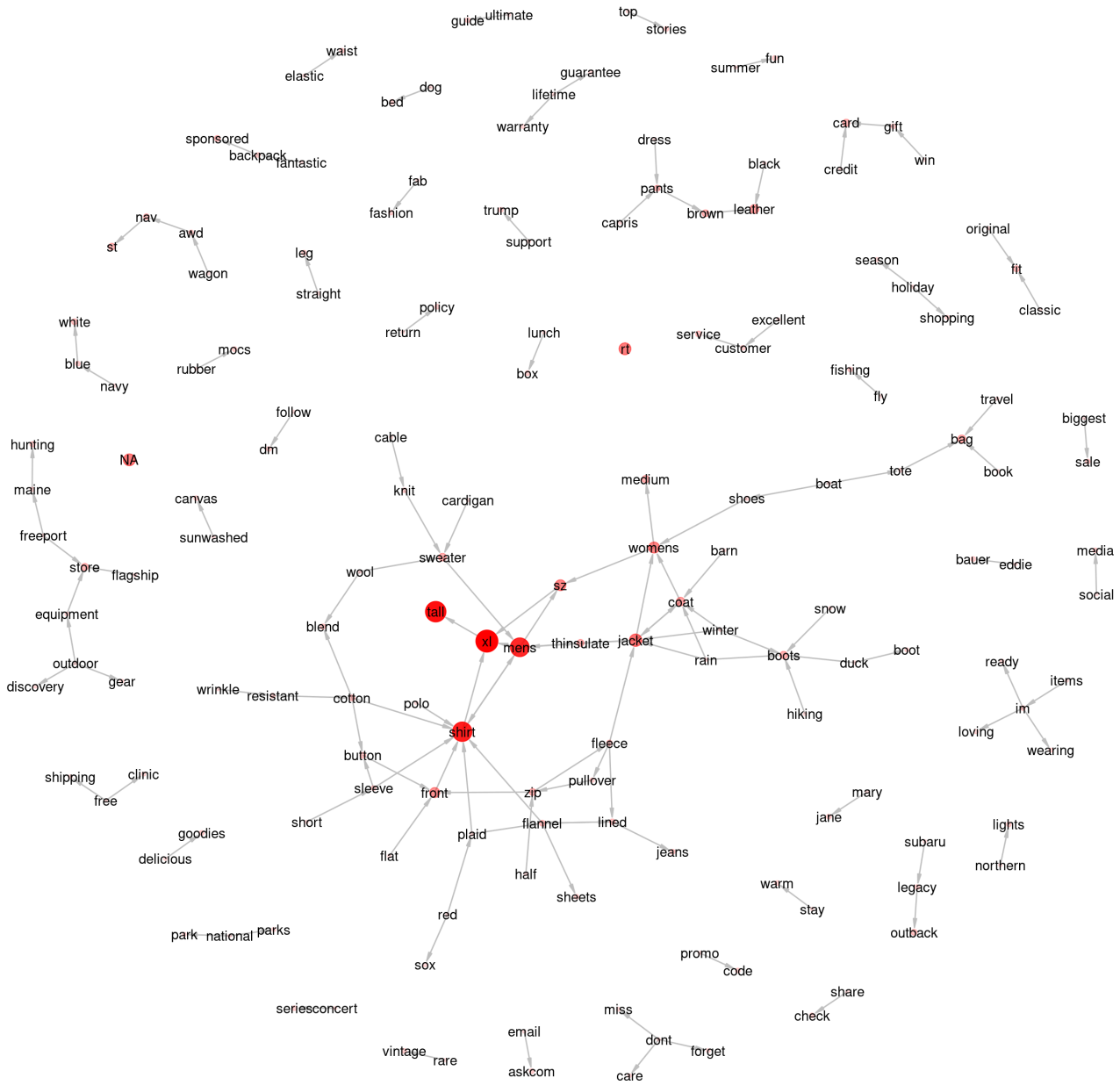
```
## [1] 139
```

```r
set.seed(2023)
# this function normalize any scale into a scale of 0 to 1
normalize <- function (x, from = range(x), to = c(0, 1)) {
  x <- (x - from[1])/(from[2] - from[1])
  if (!identical(to, c(0, 1))) {
    x <- x * (to[2] - to[1]) + to[1]
  }
  x
}

# create the graph for top bigrams and add popularity of node
bigram_graph <- as_tbl_graph(LLBean_filtered_SNA_count_top) %>%
    # calculate pagerank centrality score as the popularity
    mutate(Popularity = centrality_pagerank())


# normalize the popularity value for better visualization effect
V(bigram_graph)$Popularity <- normalize(V(bigram_graph)$Popularity, to = c(1, 8))

# create semantic network visualization
ggraph(bigram_graph, "fr") +
  geom_edge_link(
    aes(end_cap = circle(node2.Popularity + 2, "pt")),
    edge_colour = "gray",
    arrow = arrow(
    angle = 10,
    length = unit(0.1, "inches"),
    ends = "last",
    type = "closed")
  ) +
  geom_node_point(
    aes(size= I(Popularity), alpha=Popularity),
    col = "red",
    show.legend = FALSE
  ) +geom_node_text(aes(label = name))+
  theme_graph()
```



Semantic Analysis

Problem 6: What valuable data information can you learn from the semantic network created above? (**after knitting the file to a html file, set the browser zoom level to 200% or even higher for better visibility**)

Hint: *Which nodes are the top hubs based on their PageRank score. Then, how are these top hubs connected with one another through some interesting paths passing through several other nodes? Also, are there other interesting paths passing through some nodes in this network?*

Your answer:[Context: This semantic network shows how L.L.Bean-related Twitter keywords connect. The most valuable words are not just the largest or most frequent, but the words that act as connection points between product categories, customer interests, and brand themes. Visual Highlights: The clearest hub is shirt, which appears to have high central importance because it connects many nearby terms, including mens, flannel, plaid, boots, red, polo, and top. Mens is another key hub because it connects apparel details like cotton, sweater, wool,

sleeve, casual, and dress. Jacket and coat also stand out because they connect outerwear terms with seasonal terms like rain, winter, snow, duck, and boot. Objective Information Summary: Using the network structure, shirt, mens, jacket, coat, and boots appear to be the most important nodes. From a betweenness perspective, these words likely matter because they sit between clusters and help link different product conversations. For example, shirt connects men's apparel to style terms like plaid and flannel, while jacket/coat connect apparel to weather and winter-use themes. From a closeness perspective, shirt and mens appear especially important because they are close to many other product and style terms. This means they can represent a broader apparel conversation quickly. From a PageRank perspective, shirt, mens, jacket, coat, and boots are likely influential because they are connected to other meaningful product nodes rather than isolated terms. Subjective Business Insights: The strongest path appears to be mens → shirt → flannel/plaid → fleece → jacket/coat → boots → winter/snow/outdoor gear. This path connects everyday apparel, style preferences, outerwear, footwear, and seasonal outdoor use. In business terms, the network suggests customers do not discuss L.L.Bean products in isolation. Instead, they connect products as part of a broader lifestyle: durable clothing, cold weather, and outdoor activity. Business Implications: L.L.Bean could use these central nodes as campaign anchors. Mens and shirt could support apparel-focused promotions, while jacket, coat, and boots could support winter and outdoor campaigns. The connected path from flannel/fleece to jacket/coat and boots also suggests strong cross-selling opportunities, such as pairing shirts with outerwear or boots with winter gear. Overall, the network shows that L.L.Bean's Twitter conversation is organized around practical apparel that connects style, durability, and seasonal outdoor use. https://chatgpt.com/share/69ef5885-c9f0-83ea-8664-60f4c07857d1 (https://chatgpt.com/share/69ef5885-c9f0-83ea-8664-60f4c07857d1)]

Discussion

Reflect on the ways in which the text mining results could contribute to the development of an enhanced social media marketing strategy for the company. Although this analysis won't be detailed here, you will have the opportunity to collaborate with your project team, allowing you to work together with your teammates to further examine this aspect and devise comprehensive marketing approaches.